



Fig. 3: One-time password (OTP)

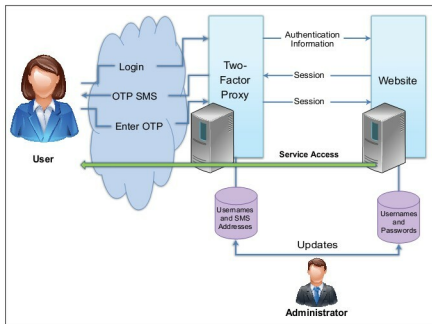


Fig. 4: Architectural view of two-factor authentication

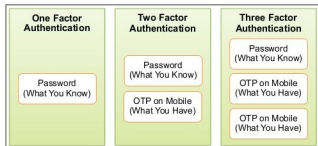


Fig. 5: Multi-factor authentication

Two-factor authentication provides an extra second layer of security with your traditional authentication system. Your account can be accessed only after successful authentication of both layers of security. Fig. 4 shows an architectural view of TFA. Two-fac-

tor authentication adds a second level of authentication—‘something you have’ like a phone OTP SMS (Fig. 3)—to the first level of authentication which is ‘something you know’ like an account login and password.

In an account login system, you have to enter the user ID and password. You can give out your user ID to anyone, but the password has to be kept a secret. In TFA system, after you enter your login details, you will get a second password or code sent

to your registered mobile phone number or e-mail address, which is known as a one-time password (OTP). There might be a third level of authentication, or a replacement of second level, where you have to be physically present during authentication. This type of authentication technique is known as ‘something you are’—like a biometric finger scanning, eye scanning or a voice print.

TFA requires a user to provide two out of three types of credentials before being able to access an account. The three types of authentication techniques (as shown in Fig. 5) are:

1. What you know—something a user knows like a password or PIN.
2. What you have—something a user owns like an ATM card or mobile phone.
3. What you are—something unique about a user like fingerprint or voice print.

What you know

Sometimes called the first level of authentication, this simple approach includes authentication of personal identification number (PIN) or password. To prevent hacking, you have to create a complex password by using some alpha and non-alpha characters including special characters. One problem you might face when using this authentication technique is that you have to remember even complex passwords, while the next authentication techniques do not need any memory power. This kind of technique is commonly known as knowledge-factor authentication, where a user has to prove the knowledge of a secret in order to authenticate.

Some examples of ‘what you know’ are:

Password. A password is a secret word or string of characters used for user authentication. This is the most commonly used mecha-